

Emergency dissemination in VANETs

AUT

Boris Gor

Supervisor: Prof. A. Al-Anbuky

Auckland University of Technology

Master of Engineering

Problem

Highly dynamic networks and frequent fragmentation prevent reliable communication in VANETs.

Emergency messages must propagate even when:

- no routes exist
- connectivity is intermittent

BGKP Protocol

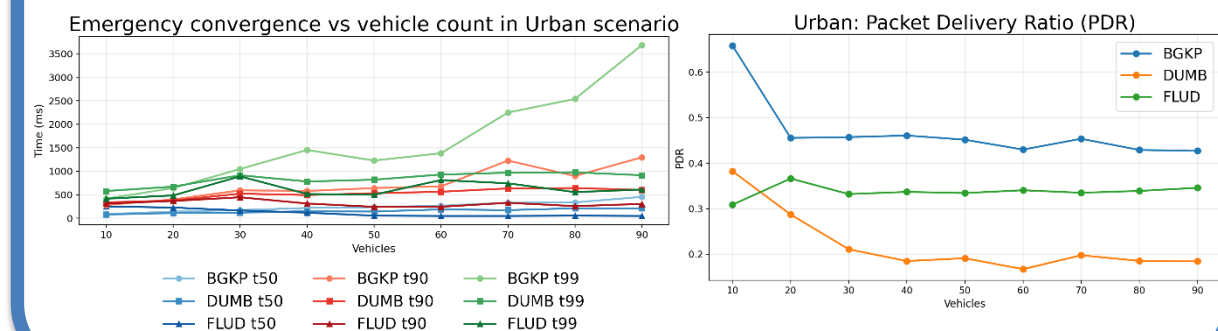
Bounded Gossip-based Kernel Protocol

- Adaptive gossip forwarding
- Bounded retransmissions
- Store-carry-forward
- Local decision making

Results

Urban Scenario (Wynyard Quarter)

- Highest urban RSU-delivered ratio
- Highest latency and transmission cost
- Emergency convergence not consistently best



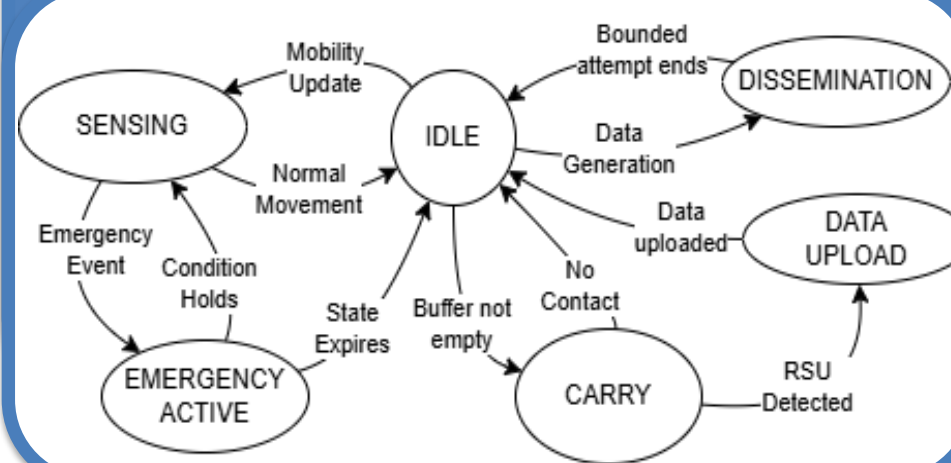
Key Idea

Infrastructure-independent emergency dissemination

Reliable notification under intermittent connectivity

- operates under network fragmentation
- does not rely on RSU availability
- supports delayed and opportunistic

System Behaviour

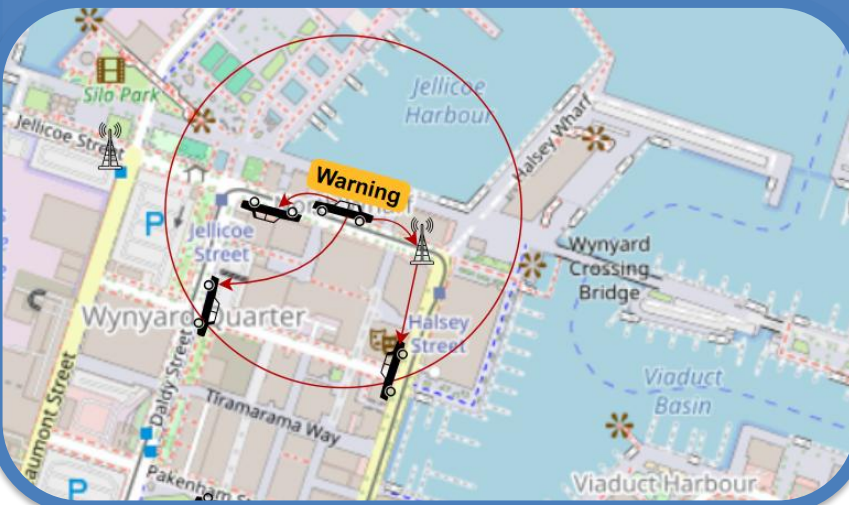


Conclusion

Trade-off design under fragmentation

- improved urban RSU delivery
- higher delay and overhead
- no consistent emergency advantage

Scenario



Key mechanisms

- Preferred peer selection
- Local decision making
- Opportunistic dissemination
- State-based emergency updates
- Bounded buffering

Methodology

- Designed BGKP and two baseline protocols
- Implemented in Contiki-NG / Cooja
- Generated urban mobility from OSM + SUMO
- Built highway mobility model in Cooja
- Processed logs into metrics and plots